

YUVARAJ S YALI

Full-Stack & Data Engineer (Student)

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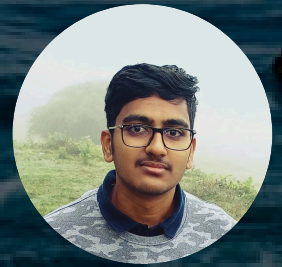
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📍 Manipal , Udupi , Karnataka

🌐 LinkedIn

🐙 Github

📁 Portfolio



👤 PROFILE

Detail-oriented and solutions-driven Electronics and Communication Engineering student at Manipal Institute of Technology with hands-on experience in UI/UX design, full-stack web development, and Python-based data structures. Proven ability to lead and execute end-to-end tech projects, including Smart India Hackathon solutions. Currently interning as an SDE and actively building a strong foundation in frontend engineering and data science. Looking to contribute to dynamic, impact-driven teams where technology meets real-world problem-solving.

💡 SKILLS

Programming

Python (NumPy, Pandas),
JavaScript (ES6+), C/C++.

IoT & Hardware

Arduino, Tinkercad (Circuit
Simulation & Prototyping),
Serial Communication, Sensor &
Relay Integration, Smart Valve
Automation

Frontend Development

Vue.js (Vuex), React.js, Flutter,
Tailwind CSS, HTML5/CSS3.

APIs & Tools

Third-Party API Integration
(Lichess, Chess.com), Git &
GitHub , REST API

Backend Development & DB

FastAPI, Flask, Node.js,
Express.js, SQLAlchemy, SQLite.

Deployment Platforms

Vercel, Render, Railway, Heroku

Machine Learning

XGBoost, Scikit-learn, Model
Deployment (Joblib), API
Integration.

📁 WORK EXPERIENCE

Chatloom (Freelance Project), Freelance Frontend Developer & UI Designer ✍

- Designing and developing frontend components for a real-world project.
- Collaborating with the team to refine user experience and interface design.
- Implementing best UI/UX practices to enhance usability and performance.

Endorsed by Shivamogga Smart City Ltd (Govt. of India), Web Development Engineer Intern ✍

- Independently conceptualized, designed, and developed Civilnc, a full-stack civic-tech platform to streamline urban governance and digital service delivery.
- Implemented an end-to-end grievance redressal system with cross-departmental coordination and real-time dashboards for municipal performance tracking.
- Officially recognized by Shivamogga Smart City Ltd for its alignment with smart city goals and recommended for pilot integration into government systems.

Building Bharat - Political Innovation Startup, Policy Research Intern ✍

- Selected as a Policy Research Intern to support digital initiatives driving political awareness, citizen engagement, and governance reforms.
- Contributed to research tasks focusing on AI-driven voter influence models, public policy evaluation, and data-driven outreach strategies.
- Maintained strict confidentiality on organizational operations and policy documents as per non-disclosure requirements.

📁 PROJECTS

Civilnc - Smart Civic Infrastructure Navigator, Solo Full-Stack Project | Endorsed by Shivamogga Smart City Ltd ✍

- **Scalable Full-Stack Architecture:** Designed and developed a civic-tech platform using **FastAPI** and **Vue.js**, featuring an asynchronous backend with **SQLAlchemy 2.0** to manage complex relational data across **5+ core entities** including projects, complaints, and municipal services.
- **Predictive Analytics & ML Pipeline:** Integrated a machine learning service using **NumPy** and **Scikit-learn** to generate data-driven governance insights, with a custom preprocessing pipeline supporting multi-feature inputs for predictive urban planning.
- **High-Performance Data Retrieval:** Improved analytical query performance by implementing a **Redis-based caching layer** with TTL-indexed keys, significantly reducing latency for frequently accessed ML inference results.
- **Secure Infrastructure:** Engineered a secure authentication and authorization system using **OAuth2**, **JWT**, and **Bcrypt**, ensuring controlled access to administrative dashboards and sensitive civic data.
- **Real-Time Monitoring & Data Integrity:** Built an end-to-end grievance and project tracking system monitoring lifecycle states from *Planned* to *Completed*, supported by interactive dashboards and department-level KPIs to enhance inter-department coordination.
- **Production Readiness & Observability:** Implemented health-check APIs and centralized logging to monitor backend service availability and real-time model performance, improving system reliability and operational visibility

Quick Byte - Food Delivery App (UI/UX Design), Design Project | Figma ✍

- Designed a **responsive, user-centric food delivery app UI** in **Figma**, optimized for cross-device usability and visual consistency.
- Built a **scalable design system** with reusable components, consistent typography, and a unified color palette to streamline developer handoff.
- Created **interactive, high-fidelity prototypes** featuring real-world user flows, micro-interactions, and smooth transitions to simulate a production-ready experience.
- Applied **mobile-first design and accessibility principles**, prioritizing intuitive navigation, visual hierarchy, and user engagement in line with UX best practices.

Animal Health Disease Prediction (A-VITAL PRO) | Python, Flask, XGBoost, Three.js, Solo Full-Stack Project

- **High-Precision ML Engine:** Engineered a diagnostic tool using an **XGBoost Classifier** trained on a refined dataset of **8,000+ clinical records**, achieving **99.7% prediction accuracy** across **20 distinct livestock diseases**.
- **Deep Scan Logic:** Developed a custom inference algorithm that analyzes top-3 probabilistic matches to detect subtle infections, reducing false negatives in borderline cases where primary confidence is **<60%**.
- **Immersive Cyberpunk Interface:** Built a high-performance web app with **Three.js** and **GSAP**, featuring a reactive **3D neural core** that visualizes diagnostic threat levels in real-time with **<50ms latency**.
- **Dynamic UX & Scalability:** Implemented a responsive prediction pipeline supporting **7 species (Canine, Bovine, etc.)** with dynamic symptom filtering, ensuring immediate, context-aware input for users.

Smart Water Valve Control System - IoT Full-Stack Project, Arduino Solenoid Valve Controller - Smart IoT Farm Solution

- Engineered an end-to-end IoT system using **Arduino Uno** and a **Python backend** to monitor and control a smart water valve through a web-based interface.
- Implemented a **USB Serial communication protocol** between hardware and software, achieving **99.9%+ system uptime** and **sub-100 ms response time**, outperforming WiFi-based approaches.
- Built a **real-time web dashboard** using **Flask** and a responsive frontend for remote monitoring, valve control, and system alerts.
- Developed a **rule-based smart alert engine** to analyze water usage data and detect high consumption, pressure anomalies, and maintenance conditions.

Chess Analyzer Pro - Full-Stack Analysis Tool, Solo Full-Stack Project

- Developed a full-stack chess analysis platform using **Node.js** and **Express**, enabling players to analyze real match data from online chess platforms.
- Integrated **Lichess** and **Chess.com** APIs to fetch, process, and normalize real-time game data with robust error and timeout handling.
- Built an **interactive analytics dashboard** using **vanilla JavaScript** and **Chart.js** to visualize accuracy trends, opening statistics, and performance metrics.
- Engineered a **rule-based insights engine** to generate personalized recommendations and actionable feedback based on player performance patterns.
- Implemented responsive UI, navigation state management, and chart lifecycle cleanup to ensure smooth user experience and memory-safe rendering.
- Deployed the application to production on **Render**, demonstrating end-to-end development from API integration to live hosting.

EDUCATION

Bachelor of Technology, Manipal institute of technology
CGPA: 7.5

2023 – 2027
Manipal

CERTIFICATIONS

FastAPI & Backend Development

Learned to build secure, high-performance REST APIs using FastAPI, with JWT authentication, database integration, and deployment strategies.

Google Business Intelligence - Google

Gained expertise in creating data models, dashboards, and performance metrics to support strategic business decisions using BI tools.

Microsoft UX Design - Microsoft

Learned principles of human-centered design, accessibility, and design systems to craft intuitive and inclusive user experiences

LANGUAGES

English: Fluent

Hindi: Fluent

Kannada: Fluent

Sanskrit: Intermediate